

advances with each clock pulse, the outputs toggle at different rates, with Q0 of U3 being the fastest and Q3 the slowest, producing the stepped waveform.

As a result, the circuit produces a stepped analogue output. The resistor network connected to the counter outputs combines the binary signals to generate a cor-

responding analogue voltage at connector J1.

As the counter advances, the output changes in discrete steps, forming a sawtooth waveform. This configuration is useful for applications such as digital-to-analogue conversion, signal generation, and basic waveform synthesis.

PARTS LIST (For circuit in Fig. 5)

Semiconductors:

- IC1 - L7805, 5V voltage regulator
- IC2 - NE555P timer IC
- IC3 - 74LS93 binary counter IC
- IC4 - LM358 op-amp
- (IC4A-IC4C) (IC4B not used)

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R7-R9 - 1k Ω
- R2, R10, R11 - 10k Ω
- R3-R6 - 2k Ω

Capacitors:

- C1 - 1 μ F, 25V electrolytic
- C2, C3, C5 - 0.1 μ F ceramic
- C4 - 0.47 μ F, 25V electrolytic

Miscellaneous:

- J1 - Output connector
- Breadboard/general-purpose PCB
- Connecting wires
- 9V battery with a clip

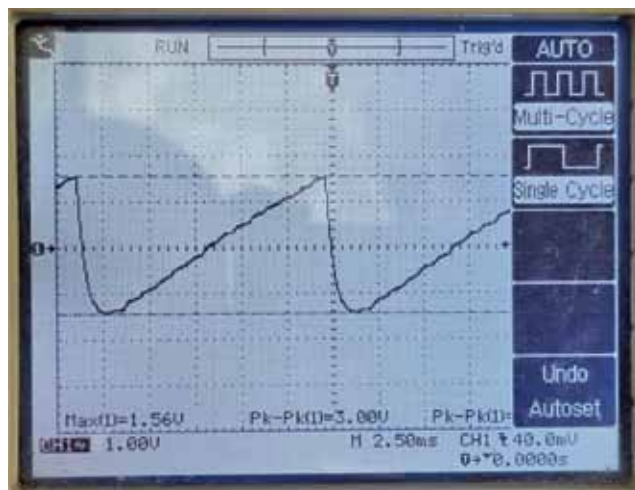


Fig. 4: A smoother waveform is obtained after connecting the smoothing capacitor

As shown in Fig. 1, no op-amp or smoothing capacitor is used; therefore, the output waveform appears as a clear staircase (see Fig. 2). A digital storage oscilloscope (DSO) displays all 16 distinct voltage

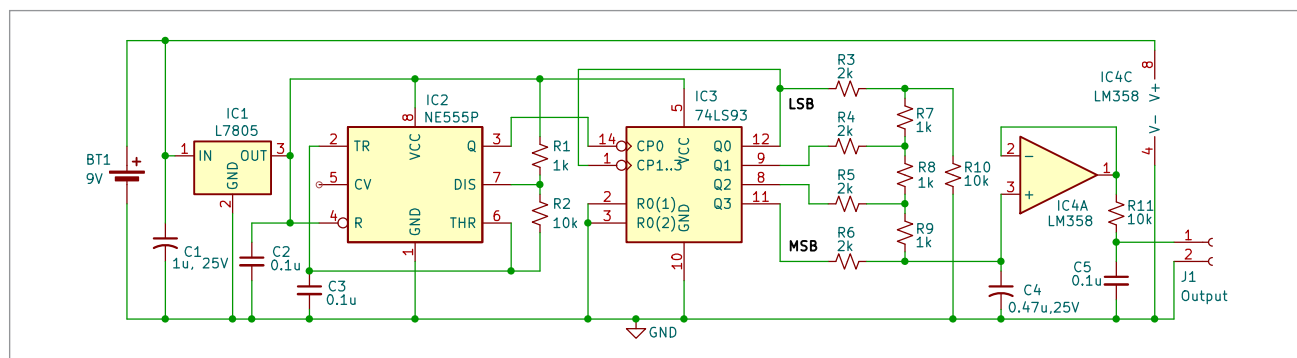


Fig. 5: Circuit diagram of sawtooth generator after adding op-amp LM358 at the output

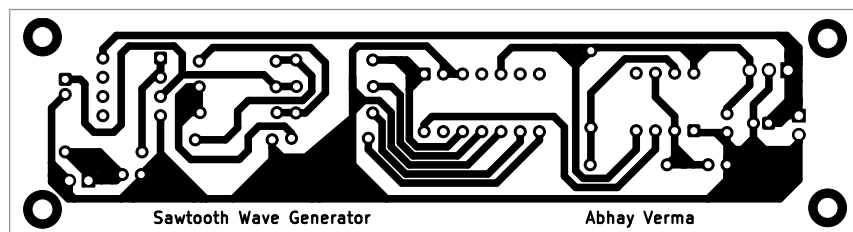


Fig. 6: Actual-size PCB of final sawtooth generator circuit

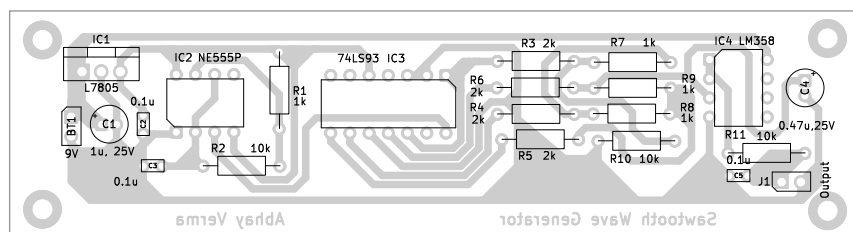


Fig. 7: Components layout

levels, confirming the correct operation of both the counter and the resistor network.

If a capacitor (C3) is connected across the output at J1, without using an op-amp, as shown in Fig. 3, the stepped waveform gets smoothed due to the capacitor's filtering effect. As a result, the output transitions from a discrete stepped signal to a more continuous waveform, as shown in Fig. 4. The DSO clearly shows that a smoother waveform is obtained after adding the smoothing capacitor across the output.

Further, if an op-amp (LM358) is added, as shown in Fig. 5, along with a smoothing capacitor C4, the